New York City Stormwater Manual

Appendix G
DETENTION IN SERIES WORKBOOK AND EXAMPLES

Environmental Protection

Detention in Series Example

A site in Queens consists of a multistory office building and a parking lot for its tenants. The site was proposed to connect to a 15 in. combined sewer. The building owner intends to use a blue roof and detention tank in series to meet the stormwater management requirement. The total roof area will be used for detention. Design a blue roof and a downstream detention system that treats runoff from the roof and the parking lot, given the following:

Total Contributing Area = 40,000 sf
Roof (sloped 1/8 in per ft) = 20,000 sf @ 0.95 runoff coefficient.
Paved = 20,000 sf @ 0.85 runoff coefficient

Use the Detention In Series Workbook provided in the accompanying Appendix G workbook Excel file.

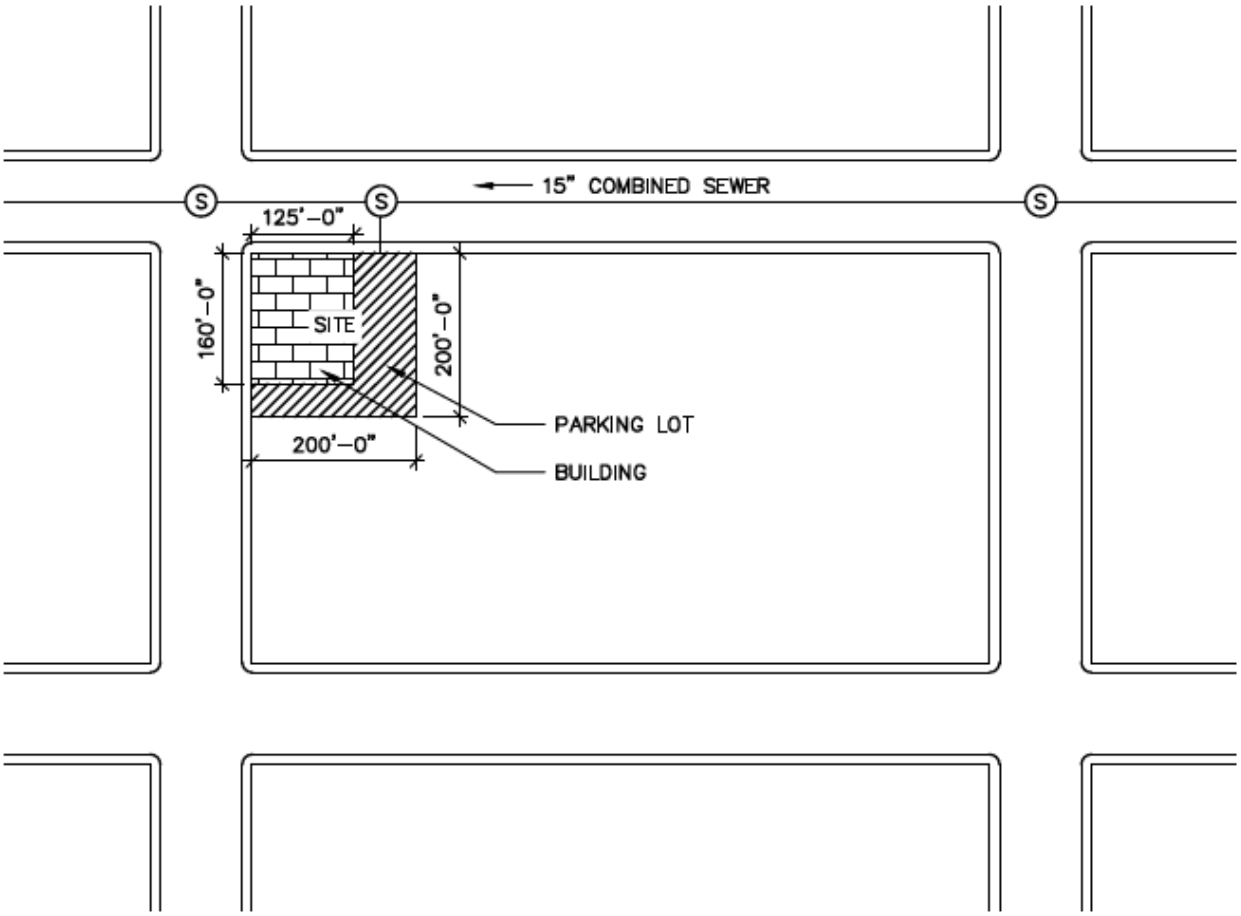

Figure G.1. Schematic of Example 1 (Not to Scale)

Step 1: Input the properties of the blue roof that will drain into the downstream detention system.

The first upstream area that drains to the downstream detention system is the 20,000 sf blue roof.

UPSTREAM SYSTEM

INPUTS					OUTPUTS	
TDA ID	TDA Area	C-value	Detention System Type	Maximum Release Rate	Required Detention Volume	Effective C-value
name	sf	#	name	cfs	cf	#
1	20000	0.95	Blue Roof			

Figure G.2. Inputs for the Blue Roof Properties

Step 2: Design the maximum release rate to be maintained by the blue roof.

Identify a controlled-flow roof drain by an approved manufacturer. In this case, the designer has selected a controlled-flow roof drain that restricts flow to 10 gpm/in. Controlled flow roof drains may have a standard flow rate per unit depth, controlled by a parabolic weir, or may have a flow rate through a custom orifice, which has different design requirements.

The roof has an area of 20,000 sf. According to the 2014 Plumbing Code by the NYC Department of Buildings, not less than four roof drains shall be installed in roofs over 10,000 sf in area. In this case, the designer has chosen to install four roof drains.

Ponding depths should not exceed 4 inches above the low point (or as specified in the current Construction Codes). The designer has chosen to use a ponding depth of 2 inches.

$$Q_{ROOF} = \frac{Q_i N_{RD} d_{max}}{449}$$

where:

Q_{ROOF} = maximum release rate from rooftop detention (cfs)

Q_i = maximum release rate from each drain (gpm/in) = 10 gpm/in

N_{RD} = number of roof drains = 4

d_R = the roof drain depth of flow (in) = 2 in

$$Q_{ROOF} = \frac{10\frac{gpm}{in} * 4 * 2\ in}{449}$$

$$Q_{ROOF} = 0.18\ cfs$$

The blue roof can maintain a maximum release rate of approximately 0.2 cfs. Input this maximum release rate into the workbook.

UPSTREAM SYSTEM

INPUTS					OUTPUTS	
TDA ID	TDA Area	C-value	Detention System Type	Maximum Release Rate	Required Detention Volume	Effective C-value
name	sf	#	name	cfs	cf	#
1	20000	0.95	Blue Roof	0.2		

Figure G.3. Input for the Maximum Release Rate Maintained by the Blue Roof

Step 3: Based on the inputs from Steps 1 and 2, the workbook will automatically calculate the duration of a storm (min) with a 10-year return frequency. This calculation is shown below.

The total roof area will be used for detention. Therefore, the available area is the entire 20,000 sf.

$$t_V = 0.27(\frac{C_{WT}A_t}{Q_{DRR}})^{0.5} - 15$$

where:

t_V = the duration of the storm with a 10 yr. return frequency requiring the maximum detention volume with a variable outflow (min)

C_{WT} = the weighted runoff coefficient for the contributing area = 0.95

A_t = contributing area (sf) = 20,000 sf

Q_{DRR} = maximum release rate for the site (cfs) = 0.2 cfs

$$t_V = 0.27(\frac{0.95 * 20{,}000\ sf}{0.2\ cfs})^{0.5} - 15$$

$$t_V = 68.2\ min$$

Step 4: Based on the inputs from Steps 1 and 2, the workbook will automatically calculate the required detention volume through the blue roof. This calculation is shown below.

$$V_V = (\frac{0.19C_{WT}A_t}{t_V + 15} - 40Q_{DRR})t_V$$

where:

V_v = the maximum required detention volume (cf)

C_{WT} = the weighted runoff coefficient for the contributing area = 0.95

A_t = contributing area (sf) = 20,000 sf

t_V = the duration of the storm with a 10 yr. return frequency requiring the maximum detention volume with a variable outflow (min) = 68.2 min

Q_{DRR} = maximum release rate for the site (cfs) = 0.2 cfs

$$V_V = [\frac{0.19 * 0.95 * 20{,}000\ sf}{68.2\ min + 15} - (40 * 0.2\ cfs)](68.2\ min)$$

$$V_V = 2{,}414\ cf$$

UPSTREAM SYSTEM

INPUTS					OUTPUTS	
TDA ID	TDA Area	C-value	Detention System Type	Maximum Release Rate	Required Detention Volume	Effective C-value
name	sf	#	name	cfs	cf	#
1	20000	0.95	Blue Roof	0.2	2414	

Figure G.4. Output for the Required Detention Volume Through the Blue Roof

Step 5: Check that the available storage volume of the roof is greater than the required detention volume.

The total roof area cannot be used for detention, because some roof area tributary to the system will be a bulkhead or parapet, and volume can't be provided in those. Therefore, the available area is set by the designer and must be less than 20,000 sf. For the purpose of this example, it will be assumed that the available area is 19,000 sf.

The designer has considered a multi-directionally sloped roof configuration, as shown in Figure G.5.

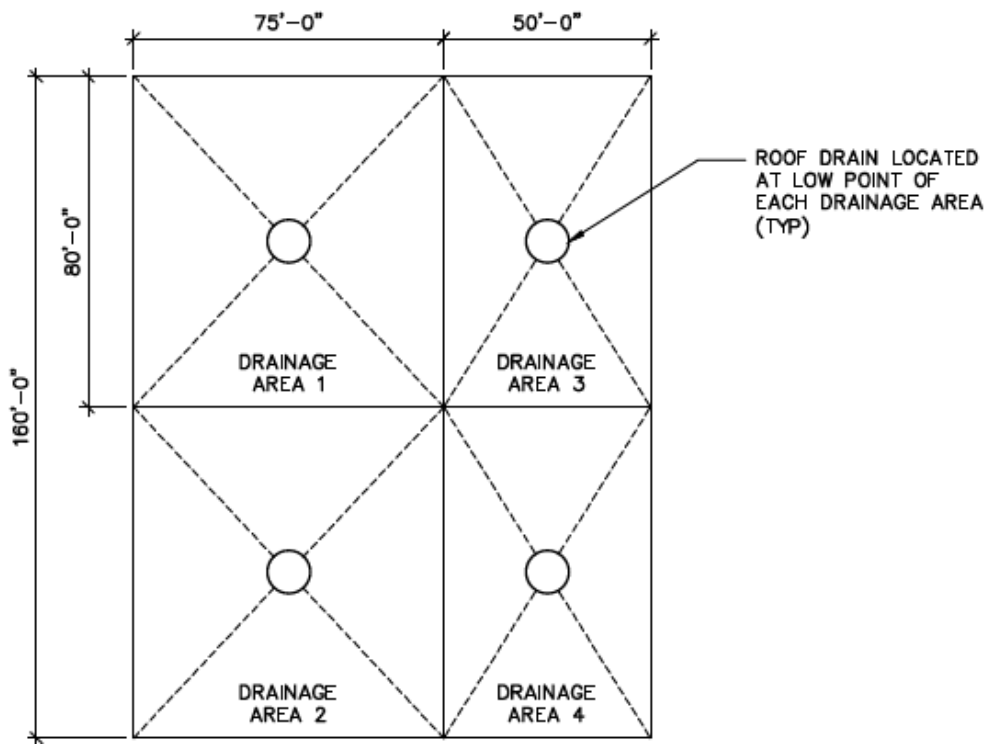

Figure G.5. Plan View of Multi-Directionally Sloped Blue Roof

To calculate the provided volume to meet the operations volume requirement, Vv, BWSO does not analyze the volumes of the tributary areas individually and assumes a best case scenario for tributary areas. The area available for detention is always less than the tributary area, and it is assumed that the area available for detention here is 18,000 sf.

For the example above, the best case scenario is reflected by determining the weighted average of the length and width of a typical drainage area, as follows, assuming a L:W ratio of 1.25:1:

Drainage Area: 75'L x 60'W

If the roof is sloped 1/8 in per ft, or about 1% the height of the short edge of an average drainage area, d_S is 3.6” above the center drain, and the depth to the long edge d_L is 4.5” above a center drain. The roof slope for roof detention should not be less than 0.5%, or 0.0625 in per ft.

Note: the height of any overflow must be between 2” and 4” above the primary drain to meet requirements from the NYC Department of Buildings.

Calculate the available storage volume of an average drainage area, accounting for the best case scenario and a constant roof slope.

There are three components to the volume provided for an average drain area, depending on the depth of flow.

1) If the depth of flow will be less than or equal to the shortest overflow height, the volume provided is calculated by checking the volume of an inverted pyramid with base width equal to the width specified above, for example 61.6’.

   Typical Drainage Area, Inverted Pyramid:

$$V_A = \frac{1}{3}W^2 * \frac{d_R}{12}$$

   where:
   V_A = the available storage volume of a typical drainage area (cf)
   W = the width of a typical drainage area (ft)
   d_R = the roof drain depth of flow (in)

2) If the depth of flow, d_R = the roof drain depth of flow (in) will be between the shortest overflow height and the longest overflow height, the volume provided includes the full inverted pyramid, calculated using 1), and adding the volume in an inverted trapezoidal prism up to the long overflow depth. The volume up to this depth is calculated with the formula:

$$V_A = \frac{1}{3}W^2 \cdot \frac{d_S}{12} + \frac{\left(16.67 \cdot \frac{d_R}{S} + W\right)}{2} W \cdot \frac{(d_R - d_S)}{12}$$

   where:
   V_A = the available storage volume of a typical drainage area (cf)
   W = the width of a typical drainage area (ft)
   L = the length of a typical drainage area (ft)
   d_R = the roof drain depth of flow (in)
   d_S = the depth to the short overflow height (in)
   S = the roof slope, in units of percentage (between 0 and 100%)

3) If the depth of flow, d_R = the roof drain depth of flow (in) will be above the longest overflow height, the volume provided includes 1), 2) and adds the volume of a rectangular prism up to the maximum overflow height.

$$V_A = \frac{1}{3}W^2 \cdot \frac{d_S}{12} + \frac{(L + W)}{2} W \cdot \frac{(d_L - d_S)}{12} + (L \cdot W)\frac{\left(d_R - (d_L + d_S)\right)}{12}$$

where:
V_A = the available storage volume of a typical drainage area (cf)
W = the width of a typical drainage area (ft)
L = the length of a typical drainage area (ft)
d_R = the roof drain depth of flow (in)
d_S = the depth to the short overflow height (in)
d_L = the depth to the long overflow height (in)

In the example, using the assumed roof drain of 10 GPM/in/weir, the depth must not exceed 2.0 inches to correctly restrict flow from the facility to 0.18 cfs. Because the depth will not exceed 2.0 inches, the first equation applies:

$$V_A = \frac{1}{3} * 60^2\ ft^2 * \frac{2.0\ in}{12} = 200\ cf$$

This indicates that only 200 cf of volume will be provided in the area of an average inverted pyramid, so with four drains the provided volume would be 800 cf, but the volume required is 2414. This demonstrates that the volume requirements are not being met by the proposed drain system, and as a result, another drain configuration is required.

Since the maximum overflow height is known to be 4" above the primary drain, it is possible to set this as the depth of flow d_R to determine a more appropriate drain. With the depth of flow not to exceed 4", and the release rate from all drains not to exceed 0.2 cfs at this depth, a target drain flow rate can be determined:

$$Q_{\frac{GPM}{in},max} = \frac{\left(Q_{drr}\,(cfs) \cdot 448\ \frac{gpm}{cfs}\right)}{\#Drains \cdot Depth\ of\ flow\ (in)}$$

In this example, the maximum flow rate for an individual drain at a depth of 4" should not exceed 5.6 GPM/inch.

With this drain, the provided volume can be calculated using the above relationships, with a flow depth of 4". This is scenario 2) above, as the flow depth is between the short overflow height and the long overflow height.

$$V_A = \frac{1}{3}(60)^2 \cdot \frac{3.6}{12} + \frac{\left(16.67 \cdot \frac{4}{1} + 60\right)}{2} 60 \cdot \frac{(4 - 3.6)}{12} = 486.7\ cf$$

For a system with four drains, this has a provided volume of ~1,946 cf, which is less than the required volume of 2,414.

Since the required detention volume is greater than the available storage volume, the user may select a different controlled-flow roof drain or design depth of flow and re-run Steps 2-4.

In this case, there is no circumstance where a drain can restrict flow sufficiently below the overflow to meet the volume requirements, so the designer has must choose a different drain, roof slope, or release

rate from the roof detention system. With a roof slope of 0.5%, the system has enough volume, but the roof slope is not always easily adjusted.

In the case when the roof slope may not be adjusted. If the flow depth is chosen to be 4 inches as the new design depth of flow. A drain that discharges at a flow rate of 5 GPM/in is selected. The new ponding depth results in a maximum release rate of 0.43 cfs, a required detention volume of 1,938 cf, and a total available storage volume of 1,947 cf.

The inputs have been updated, and the workbook automatically outputs the new required detention volume of 1,938 cf.

UPSTREAM SYSTEM						
INPUTS					OUTPUTS	
TDA ID	TDA Area	C-value	Detention System Type	Maximum Release Rate	Required Detention Volume	Effective C-value
name	sf	#	name	cfs	cf	#
1	20000	0.95	Blue Roof	0.43	1938	

Figure G.6. Inputs and Output for the Required Detention Volume Through the Blue Roof, Using a Ponding Depth of 3"

Step 6: Based on the inputs from Steps 1 and 2, the workbook will automatically calculate the effective weighted runoff coefficient for the blue roof. This calculation is shown below.

$$C_{WE} = \frac{311 Q_{DRR}(t_V + 15)}{A_t}$$

where:

C_{WE} = the effective weighted runoff coefficient for the roof with runoff restricted by controlled-flow roof drains

Q_{DRR} = maximum release rate for the site (cfs) = 0.27 cfs

t_V = the duration of the storm with a 10 yr. return frequency requiring the maximum detention volume with a variable outflow (min) = 56.6 min

A_t = contributing area (sf) = 20,000 sf

$$C_{WE} = \frac{311 * 0.43\ cfs * (56.6\ min + 15)}{20{,}000\ sf}$$

$$C_{WE} = 0.380$$

UPSTREAM SYSTEM						
INPUTS					OUTPUTS	
TDA ID	TDA Area	C-value	Detention System Type	Maximum Release Rate	Required Detention Volume	Effective C-value
name	sf	#	name	cfs	cf	#
1	20000	0.95	Blue Roof	0.43	1938	0.38

Figure G.7. Output for the Effective C-Value of the Blue Roof

Step 7: Input the properties of the parking lot that will drain into the downstream detention system.

The second upstream area that drains to the downstream detention system is the 20,000 sf parking lot. Since there is no detention system specifically for the parking lot, the effective weighted runoff coefficient remains as 0.85. The workbook will automatically output this value.

UPSTREAM SYSTEM						
INPUTS					OUTPUTS	
TDA ID	TDA Area	C-value	Detention System Type	Maximum Release Rate	Required Detention Volume	Effective C-value
name	sf	#	name	cfs	cf	#
1	20000	0.95	Blue Roof	0.43	1938	0.38
2	20000	0.85	None			0.85

Figure G.8. Inputs and Output for the Parking Lot

Step 8: Calculate the release rate to be maintained by the controlled-flow orifice for the downstream detention system. Use the maximum release rate per acre (q) shown in Table 2.9, Chapter 2.

Since the project is 20,000 sf or more, and consists of a multistory office building, this project requires a site connection permit (SCP). In addition, the site is connecting to a 15 in. combined sewer.

Table 2.9. Maximum release rate per acre (cfs/acre) by sewershed type.

q (cfs/acre)	Description
1.0	MS4 areas
0.1	CSS areas

According to Table 2.9, $q = 0.1\,\frac{cfs}{acre}$.

$$Q_{DRR} = \frac{q\ *A}{43560}\ or\ 0.046\ \ [whichever\ is\ greater]$$

where:

Q_{DRR} = maximum release rate for the site (cfs)

q = maximum release rate per acre (cfs/acre) = 0.1 cfs/acre

A = contributing area (sf) = 40,000 sf

$$Q_{DRR} = 0.092\ cfs\ > 0.046\ cfs$$

The maximum release rate is 0.092 cfs.

Step 9: Input the properties of the downstream detention system. Use the maximum release rate from Step 8.

Since the project is 20,000 sf or more, and consists of a multistory office building, this project requires a site connection permit (SCP). The site has a total contributing area of 40,000 sf.

DOWNSTREAM SYSTEM

INPUTS			**OUTPUTS**	
Permit Type	Total Contributing Area	Maximum Release Rate	Required Detention Volume	Effective C-value
name	sf	cfs	cf	#
CSS - SCP	40000	0.092		

Figure G.9. Inputs for the Downstream Detention System

Step 10: Based on the inputs from Step 9, the workbook will automatically calculate the effective weighted runoff coefficient for the downstream detention system. This calculation is shown below.

$$C_W = \frac{(C_1A_1 + C_2A_2 + \cdots etc.)}{A_t}$$

where:

C_W = weighted runoff coefficient relating peak rate of rainfall and runoff

C_1 = the effective weighted runoff coefficient for the area classified as roof = 0.3[0]8

A_1 = the area classified as roof (sf) = 20,000 sf

C_2 = the effective weighted runoff coefficient for the area classified as paved = 0.85

A_2 = the area classified as paved (sf) = 20,000 sf

A_t = contributing area (sf) = 40,000 sf

$$C_W = \frac{(0.38 * 20{,}000\ sf) + (0.85 * 20{,}000\ sf)}{40{,}000\ sf}$$

$$C_W = 0.61$$

DOWNSTREAM SYSTEM				
INPUTS			OUTPUTS	
Permit Type	Total Contributing Area	Maximum Release Rate	Required Detention Volume	Effective C-value
name	sf	cfs	cf	#
CSS - SCP	40000	0.092		0.61

Figure G.10. Output for the Effective C-Value of the Downstream Detention System

Step 11: Identify the rainfall depth (R_D) based on the sewershed type and connection proposal type for the project. Use Table 2.7 in Chapter 2.

Since the project is 20,000 sf or more, and consists of a multistory office building, this project requires a site connection permit (SCP). In addition, the site is connecting to a 15 in. combined sewer.

Table 2.7. Applied rainfall depth by sewershed type and connection proposal type.

R_D	Description
1.85	CSS areas with SCP
1.50	CSS areas with HCP
1.50	MS4 areas wiith SCP
1.10	MS4 areas with HCP

According to Table 2.7, $R_D = 1.85\ in.$

Step 12: Based on the inputs from Step 9, the workbook will automatically calculate the required detention volume through the detention tank. This calculation is shown below.

$$V_V = \frac{R_D}{12} * A * C_W$$

where:

V_V = the maximum required detention volume (or sewer operations volume) (cf)

R_D = rainfall depth (in) = 1.85 in

A = contributing area (sf) = 40,000 sf

C_W = weighted runoff coefficient relating peak rate of rainfall and runoff = 0.[575]61

$$V_V = \frac{1.85\ in}{12} * 40{,}000\ sf * 0.61$$

$$V_V = 3{,}791\ cf$$

DOWNSTREAM SYSTEM				
INPUTS			OUTPUTS	
Permit Type	Total Contributing Area	Maximum Release Rate	Required Detention Volume	Effective C-value
name	sf	cfs	cf	#
CSS - SCP	40000	0.092	3791	0.61

Figure G.11. Output for the Required Detention Volume Through the Downstream Detention System

Step 13: Use the controlled-flow orifice equation to determine an appropriate orifice area for the detention tank, by assuming the active storage depth.

In order to minimize the area required for the detention tank, choose the maximum depth that is still feasible according to site limitations and use a re-entrant orifice. In this case, the designer has chosen an active storage depth of 4 ft.

$$Q_O = C_D * A_o * \sqrt{2gH}$$

where:

Q_O = maximum release rate of orifice (cfs) = 0.092 cfs

C_D = coefficient of discharge, 0.52 for re-entrant orifice

A_O = area of orifice (sf)

g = acceleration due to gravity, 32.2 (ft/s^2)

H = maximum hydraulic head above the centerline of the orifice (ft) = 4 ft

$$0.092\ cfs = 0.52 * A_o * \sqrt{2 * 32.2\left(\frac{ft}{s^2}\right) * 4\ ft}$$

$$A_o = 0.011\ sf$$

Step 14: Translate the area of the controlled-flow orifice (A_O) into a diameter and check that it is greater than the minimum diameter of 1 in.

$$A_o = \frac{\left[\pi * \left(\frac{D_o}{2}\right)^2\right]}{144}$$

where:

A_O = area of orifice (sf) = 0.011 sf

D_O = diameter of orifice (in)

$$0.011\ sf = \frac{\left[\pi * \left(\frac{D_O}{2}\right)^2\right]}{144}$$

$$D_O = 1.42\ in\ > 1\ in \qquad OK$$

Set the orifice diameter to the nearest 0.25-inch interval rounding down, with a minimum orifice diameter of one-inch. In this case, use an orifice diameter of 1.25 inches.

Step 15: Confirm the orifice area of the selected orifice diameter from Step 14.

$$A_o = \frac{\left[\pi * \left(\frac{D_o}{2}\right)^2\right]}{144}$$

where:
A_O = area of orifice (sf)
D_O = diameter of orifice (in) = 1.25 inches

$$A_O = \frac{\left[\pi * \left(\frac{1.25\ in}{2}\right)^2\right]}{144}$$

$$A_O = 0.009\ sf$$

Step 16: Confirm the required active storage depth in the tank using the orifice area from Step 15.

Compute the maximum storage depth in ft. of a detention facility with a Re-entrant orifice tube outlet, S_{DR}, with a C_D of 0.52, by the equation:

$$S_{DR} = 1{,}930\ (Q_{DRR})^2/(d_o)^4 + d_o/24$$

where:
S_{DR} = the maximum storage depth in ft. For a Re-entrant orifice tube outlet
Q_{DRR} = the detention facility maximum release rate in cfs.
d_O= the nominal dia. of the orifice tube outlet in in.

$$S_{DR} = 1{,}930\ (0.092)^2/(1.25)^4 + 1.25/24$$

$$H = 6.74\ ft$$

If the active storage depth is too high, then increase the orifice size by 0.25 inches and re-run Steps 13-14 until a suitable depth is identified. If the active storage depth is too low, then decrease the orifice size by 0.25 inches (but not less than 1 inch) and re-run Steps 13-14. Alternatively, the designer can choose a different orifice configuration as needed to modify the active storage depth.

In this case, the depth is too high to drain via gravity connection to the storm sewer. Using a flush orifice, which has a coefficient of discharge of 0.61, results in an active storage depth of 4.91 ft.

Step 17: Set the dimensions of the detention tank's active storage zone.

Based on the active storage depth of 4.91 ft and the V_V of 3,791 cf, set the interior detention tank dimensions to L: 28.0 ft and W: 20 ft. The resulting detention tank has an active storage volume of 3,849.44 cf. Note that the exterior dimensions of the detention tank will be larger than the dimensions of the active storage zone (28.0'L x 28.0'W x 4.91'D) to accommodate wall thickness, bypass structures, and/or other internal features.